

```
pip install openai-whisper
```

Collecting openai-whisper

Downloading openai_whisper-20250625.tar.gz (803 kB)

803.2/803.2 kB 18.7 MB/s eta 0:00:00

Installing build dependencies ... done

Getting requirements to build wheel ... done

Preparing metadata (pyproject.toml) ... done

Requirement already satisfied: more-itertools in /usr/local/lib/python3.12/dist-packages (from openai-whisper)

Requirement already satisfied: numba in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (0.59.0)

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (2.0.2)

Requirement already satisfied: tiktoken in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (0.8.0)

Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (2.5.1)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (4.66.1)

Requirement already satisfied: triton>=2 in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (2.40.0)

Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (0.43.0dev0)

Requirement already satisfied: regex in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (2024.5.10)

Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from openai-whisper) (2.32.0)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (3.13.1)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (4.10.0)

Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (75.0.0)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (1.13.3)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (3.4.1)

Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (3.1.3)

Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (2024.9.0)

Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.8.1)

Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.9.4)

Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (9.19.0.56)

Requirement already satisfied: nvidia-cusparse-cu12==12.8.0.53 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.8.0.53)

Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (2.28.9)

Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (3.4.5)

Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.8.4.1)

Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.8.90)

Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (11.3.3.83)

Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (1.13.1.3)

Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.8.90)

Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (10.3.9.90)

Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (11.7.3.90)

Requirement already satisfied: nvidia-cusparselt-cu12==12.5.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.5.8.93)

Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.8.93)

Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.8.93)

Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (12.8.90)

Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dist-packages (from torch->openai-whisper) (1.1)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy->openai-whisper) (1.1.0)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2->openai-whisper) (2.0.1)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->openai-whisper) (3.3.2)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->openai-whisper) (3.10)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->openai-whisper) (2.2.3)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->openai-whisper) (2024.7.4)

Building wheels for collected packages: openai-whisper

Building wheel for openai-whisper (pyproject.toml) ... done

Created wheel for openai-whisper: filename=openai_whisper-20250625-py3-none-any.whl size=803979 sha256=803979

Stored in directory: /root/.cache/pip/wheels/61/d2/20/09ec9bef734d126cba375b15898010b6cc28578d8afdde58

Successfully built openai-whisper

Installing collected packages: openai-whisper

Successfully installed openai-whisper-20250625

```
import torch
print(torch.cuda.is_available())
```

True

```
import whisper

model = whisper.load_model("medium")
```


Question: How much cement is required for a 3 BHK apartment?*

Answer: Typically **350 to 400 bags** of cement.

Question: Which cement brand do you recommend?*

Answer: It depends on project budget. Commonly used brands are **Ambuja Cement** and **UltraTech Ceme

```
chunks = text.split("Question:")
```

```
chunks = [chunk.strip() for chunk in chunks if chunk.strip()]
```

```
chunks = [f"Question: {chunk}" for chunk in chunks]
```

```
print(chunks[0])
```

Question:

```
print("Total chunks:", len(chunks))
```

Total chunks: 17

```
from sentence_transformers import SentenceTransformer
```

```
embedding_model = SentenceTransformer("all-MiniLM-L6-v2")
vectors = embedding_model.encode(chunks)
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:112: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.

To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/s>)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
```

```
modules.json: 100% 349/349 [00:00<00:00, 24.8kB/s]
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Ple

```
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 7.83kB/s]
```

```
README.md: 100% 10.5k/10.5k [00:00<00:00, 717kB/s]
```

```
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.08kB/s]
```

```
config.json: 100% 612/612 [00:00<00:00, 61.5kB/s]
```

```
model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 62.3MB/s]
```

```
Loading weights: 100% 103/103 [00:00<00:00, 3246.17it/s]
```

```
tokenizer_config.json: 100% 350/350 [00:00<00:00, 39.3kB/s]
```

```
vocab.txt: 100% 232k/232k [00:00<00:00, 8.57MB/s]
```

```
tokenizer.json: 100% 466k/466k [00:00<00:00, 22.1MB/s]
```

```
special_tokens_map.json: 100% 112/112 [00:00<00:00, 11.3kB/s]
```

```
config.json: 100% 190/190 [00:00<00:00, 21.5kB/s]
```

```
!pip install qdrant-client
```

Collecting qdrant-client

Downloading qdrant_client-1.18.0-py3-none-any.whl.metadata (11 kB)

Requirement already satisfied: grpcio>=1.41.0 in /usr/local/lib/python3.12/dist-packages (from qdrant-cl

Requirement already satisfied: httpx>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from httpx[http

```
Requirement already satisfied: numpy>=1.26 in /usr/local/lib/python3.12/dist-packages (from qdrant-client)
Collecting portalocker<4.0,>=2.7.0 (from qdrant-client)
  Downloading portalocker-3.2.0-py3-none-any.whl.metadata (8.7 kB)
Requirement already satisfied: protobuf>=3.20.0 in /usr/local/lib/python3.12/dist-packages (from qdrant-client)
Requirement already satisfied: pydantic!=2.0.*,!=2.1.*,!=2.2.0,>=1.10.8 in /usr/local/lib/python3.12/dist-packages (from qdrant-client)
Requirement already satisfied: urllib3<3,>=1.26.14 in /usr/local/lib/python3.12/dist-packages (from qdrant-client)
Requirement already satisfied: typing-extensions~=4.12 in /usr/local/lib/python3.12/dist-packages (from qdrant-client)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.20.0->httpx)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.20.0->httpx)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.20.0->httpx)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx>=0.20.0->httpx)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpcore)
Requirement already satisfied: h2<5,>=3 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpcore)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from pydantic!=2.0.*,!=2.1.*,!=2.2.0,>=1.10.8->pydantic)
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from pydantic!=2.0.*,!=2.1.*,!=2.2.0,>=1.10.8->pydantic)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic!=2.0.*,!=2.1.*,!=2.2.0,>=1.10.8->pydantic)
Requirement already satisfied: hyperframe<7,>=6.1 in /usr/local/lib/python3.12/dist-packages (from h2<5,>=3->h2)
Requirement already satisfied: hpack<5,>=4.1 in /usr/local/lib/python3.12/dist-packages (from h2<5,>=3->h2)
Downloading qdrant_client-1.18.0-py3-none-any.whl (398 kB)
 398.1/398.1 kB 13.8 MB/s eta 0:00:00
Downloading portalocker-3.2.0-py3-none-any.whl (22 kB)
Installing collected packages: portalocker, qdrant-client
Successfully installed portalocker-3.2.0 qdrant-client-1.18.0
```

```
from qdrant_client import QdrantClient

client = QdrantClient(
    url="https://a54e70c2-fbbc-4bec-bf34-71add3524d10.us-east-2-0.aws.cloud.qdrant.io",
    api_key="eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJhbnVudCI6ImFwaS1rZXk6NWVjZDhY"
)
```

```
from qdrant_client.models import Distance, VectorParams

client.create_collection(
    collection_name="construction_qa",
    vectors_config=VectorParams(
        size=384,
        distance=Distance.COSINE
    )
)
```

```
-----
UnexpectedResponse                                Traceback (most recent call last)
/tmp/ipykernel_8765/2475327263.py in <cell line: 0>()
    1 from qdrant_client.models import Distance, VectorParams
    2
----> 3 client.create_collection(
    4     collection_name="construction_qa",
    5     vectors_config=VectorParams(

5 frames
/usr/local/lib/python3.12/dist-packages/qdrant_client/http/api_client.py in send(self, request, type_)
   128     except ValidationError as e:
   129         raise ResponseHandlingException(e)
--> 130     raise UnexpectedResponse.for_response(response)
   131
   132     def send_inner(self, request: Request) -> Response:

UnexpectedResponse: Unexpected Response: 409 (Conflict)
Raw response content:
b'{"status":{"error":"Wrong input: Collection `construction_qa` already exists!"},"time":0.022390328}'
```

Next steps: [Explain error](#)

```
client.upsert(
    collection_name="construction_qa",
    points=points
)
```

```
UpdateResult(operation_id=3, status=<UpdateStatus.COMPLETED: 'completed'>)
```

```
payload = {
    "text": chunk,
    "topic": "tiles",
    "category": "costing",
    "phase": "finishing"
}
```

```
payload = {
    "text": chunk,
    "topic": "cement",
    "category": "materials",
    "phase": "foundation"
}
```

```
from qdrant_client.models import PointStruct

points = []

for idx, chunk in enumerate(chunks):
    points.append(
        PointStruct(
            id=idx,
            vector=vectors[idx].tolist(),
            payload={"text": chunk}
        )
    )
```

```
collections = client.get_collections()

for collection in collections.collections:
    print(collection.name)
```

```
book
books
construction_qa
recursive_books
sliding_books
```

```
client.count("construction_qa")
```

```
CountResult(count=17)
```

```
!pip install groq
```

```
Collecting groq
  Downloading groq-1.4.0-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: anyio<5,>=3.5.0 in /usr/local/lib/python3.12/dist-packages (from groq) (4)
Requirement already satisfied: distro<2,>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from groq) (1.7.0)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from groq) (0.23.0)
Requirement already satisfied: pydantic<3,>=1.9.0 in /usr/local/lib/python3.12/dist-packages (from groq) (1.9.0)
Requirement already satisfied: sniffio in /usr/local/lib/python3.12/dist-packages (from groq) (1.3.1)
Requirement already satisfied: typing-extensions<5,>=4.14 in /usr/local/lib/python3.12/dist-packages (from groq) (4.14)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.12/dist-packages (from anyio<5,>=3.5.0) (3.10)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0) (2024.7.4)
```

```
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from p
Requirement already satisfied: pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages (from py
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from
Downloading groq-1.4.0-py3-none-any.whl (143 kB)
```

143.7/143.7 kB 4.9 MB/s eta 0:00:00

```
Installing collected packages: groq
Successfully installed groq-1.4.0
```

```
from groq import Groq

client_llm = Groq(api_key="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")

response = client_llm.chat.completions.create(
    model="llama-3.3-70b-versatile",
    messages=[
        {"role": "user", "content": "What is RAG?"}
    ]
)

print(response.choices[0].message.content)
```

RAG can refer to different things depending on the context. Here are a few possible meanings:

1. ****RAG (Raise and Give)****: In the UK, RAG is a common abbreviation for "Raise and Give," which refers
2. ****RAG (Red, Amber, Green)****: In project management, RAG is a status reporting system used to indicate
 - * Red: The project is at risk or has significant issues.
 - * Amber (or Yellow): The project is proceeding with some caution or minor issues.
 - * Green: The project is on track and progressing smoothly.
3. ****RAG (Residual Activation Gradient)****: In artificial intelligence and machine learning, RAG is a tec
4. ****Other meanings****: RAG can also be an abbreviation for other terms, such as "Rag" (a type of fabric

Without more context, it's difficult to determine which meaning is most relevant. If you have any additi

```
query = "for plastering what should i use?"

# Generate embedding for the query
query_vector = embedding_model.encode(query).tolist()

# Retrieve relevant chunks from Qdrant
results = client.query_points(
    collection_name="construction_qa",
    query=query_vector,
    limit=17
)

# Combine retrieved chunks into context
context = "\n\n".join(
    [point.payload["text"] for point in results.points]
)

print("Retrieved Context:\n")
print(context)
```

6. Erection of columns and slabs (superstructure)
7. Brickwork / blockwork
8. Installation of electrical conduits in slabs and walls
9. Wet plaster (internal and external)
10. Finishing items: door frames, doors, windows, tiles, painting, etc.

Finishing items like tiles can cost anywhere from ₹200 per tile up to ₹5,000 or even ₹10,000 per tile,

Question: What quality standards and safety measures are followed?*

Answer: - **Quality:** Only 53-grade cement is used; TMT steel bars are standard. Preferred brands: UltraTech, Shree Cement, etc.
- **Safety:** Barricading around the site, safety belts for workers at height, helmets, and safety shoes.

Question: Any advice for first-time buyers (buying a flat)?*

Answer: Look at the surroundings – proximity to shopping malls, railway stations, schools, hospitals, and parks.

Question: What is the cost of interior work?*

Answer: Interior costs vary widely. Tiles alone can range from ₹200 per piece to ₹10,000 per piece.

Question: How much cement is required for a 3 BHK apartment?*

Answer: Typically 350 to 400 bags of cement.

Question: How much steel is required for a 3 BHK apartment?*

Answer: Approximately 300 kg to 400 kg of TMT steel bars.

Question: What additional works come under extra costing?*

Answer: Electrical, plumbing, flooring, door frames, doors, windows, and similar items. These are not included in the basic construction cost.

Question: How much time does interior work take?*

Answer: For a standard apartment, finishing work (including basic interior elements) needs at least 2-3 weeks.

Question: What is the average cost for a 2 BHK and a 3 BHK apartment?*

Answer: - **3 BHK** (1,200–1,500 sq ft): ₹25 lakh to ₹45 lakh.
- **2 BHK** (800–1,000 sq ft, typical size): Based on the same rate (approx. ₹2,100–3,000/sq ft), it falls between ₹16.8 lakh and ₹22.5 lakh.
- **Villa:** Depends heavily on size and finish. Typically ₹1,500–3,000/sq ft. A 1,200 sq ft villa could range from ₹18 lakh to ₹36 lakh.

Question: What is the typical size of a 3 BHK apartment—1,200 sq ft or 1,500 sq ft?*

Answer: Both are common. It depends on the floor plan, number of units per floor, and total built-up area.

Question: What factors affect the overall project cost?*

Answer: Primarily **location**:
- **Rural areas** – costs are lower due to cheaper labour and locally available materials.
- **Cities** – costs rise because of transportation charges and higher labour rates.

Question: How much time does it take to construct a 3 BHK apartment?*

Answer: It depends on site conditions. If work is expedited, the main structure can be completed in 3-4 months.

Question:

```
from groq import Groq

client_llm = Groq(api_key="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")

prompt = f"""
You are a construction domain assistant.

Answer the question using only the provided context.

If the answer is not present in the context, say:
"I do not have enough information."

Context:
{context}

Question:
{query}

Answer:
"""
```

```
response = client_llm.chat.completions.create(  
    model="llama-3.3-70b-versatile",  
    messages=[  
        {  
            "role": "user",  
            "content": prompt  
        }  
    ],  
    temperature=0.2  
)  
  
print(response.choices[0].message.content)
```

For plastering, you should use **River sand**. The required quantity is about **20 to 25 units**.